

AAIndex Explorer

A complete guide to exploring, analysing, and visualising amino acid physicochemical properties.

Data source: AAIndex database — Kawashima & Kanehisa, Nucleic Acids Research, 2000

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1 Overview

AAIndex Explorer is a browser-based interface for the AAIndex database — a curated collection of numerical indices representing physicochemical and biochemical properties of amino acids, as well as substitution matrices for sequence analysis.

The database contains three sub-databases:

- AAIndex1** 566 numerical indices, one value per amino acid (e.g. hydrophobicity, molecular weight, helix propensity).
- AAIndex2** 94 amino acid mutation matrices used in pairwise sequence alignment scoring.
- AAIndex3** 47 statistical protein contact potentials representing residue–residue interaction preferences.

The application is divided into six main sections accessible from the top navigation bar: **Explorer**, **Sequence**, **Encode**, **Compare**, **Visualiser**, and **About**.

2 Explorer — Browse & Search Records

The Explorer is the main entry point. It lists all records in the active database and provides tools for search, filtering, favouriting, and adding records to the comparison queue.

SWITCHING DATABASES

- 1 Click the database switcher at the top of the Explorer sidebar (AAIndex1 / AAIIndex2 / AAIIndex3).
- 2 The record list updates immediately. All search and filter state resets when you switch.

SEARCHING RECORDS

- 1 Type any keyword into the search bar — the search runs against accession codes, descriptions, and categories simultaneously.
- 2 Results are ranked by relevance using fuzzy matching, so partial or approximate terms still return useful results.
- 3 Clear the search bar to return to the full list.

Tip: Search for property keywords like "hydrophobicity", "helix", "charge", or "volume" to find related records across all categories.

FILTERING BY CATEGORY

- 1 Use the Category dropdown in the sidebar to restrict the list to a single property category (e.g. "Energy transfer", "Hydrophobicity").
- 2 Category filtering combines with the text search — both filters apply simultaneously.
- 3 Select "All categories" to remove the filter.

FAVOURITING RECORDS

- 1 Click the star icon (☆) on any record card to add it to your favourites.
- 2 Click the filled star (★) to remove it.
- 3 Enable "Show favourites only" in the sidebar to view only your starred records.
- 4 Favourites persist across sessions using your browser's local storage.

VIEWING A RECORD IN DETAIL

- 1 Click the accession code or title of any record card to open its detail page.
- 2 The detail page shows: the full description, reference, PMID, correlation coefficients with similar records, and a bar chart of values across all 20 amino acids.
- 3 Click "PubMed" to open the source publication in a new tab.
- 4 Use the "Similar records" section to discover correlated or anti-correlated properties.

ADDING RECORDS TO COMPARE

- 1 Click the "Add to compare" button on any record card.

- 2 A count badge appears on the Compare nav link showing how many records are queued.
- 3 You can queue up to 4 records at a time.
- 4 A drawer appears at the bottom of the screen showing queued records — click "Compare" to proceed.

3 Sequence Analysis

The Sequence Analysis page encodes a single protein sequence against one or more AAIndex1 properties and visualises the result as a line chart or colour-coded heatmap. It contains three tabs.

TAB 1 — SEQUENCE ENCODER

Maps each residue in a sequence to its numeric property value and plots the result as a line chart.

- 1 Paste a protein sequence (single-letter code, whitespace is stripped) into the sequence input box.
- 2 Select an AAIndex1 record from the list on the left — use the filter input to narrow by accession or description.
- 3 The line chart updates in real time. Residues with no data appear as gaps.
- 4 A colour-coded strip below the chart shows each residue letter for sequences up to 100 residues.
- 5 Click “↓ CSV” to download the per-position values as a spreadsheet.

Tip: Unknown characters (e.g. ambiguous codes like “B” or “X”) are flagged with a warning and rendered as gaps in the chart.

TAB 2 — SLIDING WINDOW

Applies a sliding window average to smooth the property profile — the classic approach for hydrophobicity plots (Kyte–Doolittle, 1982).

- 1 Paste your sequence and select a property as above.
- 2 Adjust the window size slider (range 3–25). The default is 9, which matches the original Kyte–Doolittle method.
- 3 At each position the displayed value is the mean of the window centred on that residue. Edge positions use a shorter window.
- 4 Export the smoothed values via the “↓ CSV” button.

Tip: Larger window sizes produce smoother curves that emphasise global trends (e.g. transmembrane domains). Smaller windows preserve local peaks.

TAB 3 — MULTI-PROPERTY HEATMAP

Encodes a sequence against multiple indices simultaneously and displays all profiles as a grid where colour represents the normalised value at each position.

- 1 Paste your sequence into the input.
- 2 Check or uncheck indices in the list to include or exclude them. Use “Top 10” for a quick selection.
- 3 Each row represents one AAIndex1 property; each column is one sequence position.

- 4 Colour runs from blue (low) through white to red (high), normalised independently per row.
- 5 Hover a cell to see the exact value and residue identity in a tooltip.

4 Encode — Batch Sequence Encoding

The Encode page processes entire FASTA files or plain-text sequence lists and encodes all sequences against a selected AAINdex1 property. It is designed for batch workflows where you need to generate feature vectors for machine learning or downstream analysis.

UPLOADING SEQUENCES

- 1 Click the upload area or drag-and-drop a file onto it. Accepted formats: .fasta, .fa, .txt.
- 2 FASTA files: sequences are separated by ">" header lines. The first word after ">" is used as the sequence ID.
- 3 Plain text files: one sequence per line. Sequences are automatically assigned IDs (seq_1, seq_2, ...).
- 4 Invalid lines with no recognisable amino acid characters are silently skipped.

Tip: Sequences are uppercased on import. Whitespace within lines is stripped. Ambiguous residues are accepted but will produce null values in the encoding.

SELECTING AN AAINDEX1 RECORD

- 1 Use the filter input in the left sidebar to search records by accession or description.
- 2 Click any record in the list to select it as the encoding property.
- 3 The preview chart and sequence table update immediately.
- 4 Your starred records appear in the Favourites panel (click the arrow to expand) for quick access.

READING THE PREVIEW

- 1 The preview panel shows a line chart of the first (or selected) sequence's encoded values.
- 2 The actual protein sequence is shown in a scrollable monospace box directly above the chart.
- 3 The sequence table lists all loaded sequences with their length and a summary of encoding statistics.
- 4 Click any row in the table to load that sequence into the preview chart.

EXPORTING

- 1 "↓ All encodings (long CSV)" — one row per residue per sequence. Columns: sequence_id, position, amino_acid, value.
- 2 "↓ Summary stats CSV" — one row per sequence. Columns: sequence_id, length, valid_residues, mean, min, max.

5 Comparator — Side-by-Side Comparison

The Comparator lets you place up to four AAIIndex1 records side by side in a table and examine how property values differ across all 20 amino acids.

BUILDING THE COMPARISON SET

- 1 Navigate to the Explorer and click “Add to compare” on each record you want to include (maximum 4).
- 2 The sticky drawer at the bottom of the screen shows queued records.
- 3 Click “Compare” in the drawer, or navigate directly to the Compare page from the navbar.
- 4 Records can also be removed from the drawer by clicking the × next to their accession.

READING THE COMPARISON TABLE

- 1 Each column represents one AAIIndex1 record; each row is one amino acid.
- 2 Toggle “Normalise (z-score)” to scale all values to comparable z-scores — useful when records use different units.
- 3 Values are colour-coded within each column: dark blue = high, dark red = low, grey = missing.
- 4 Click any column header to sort by that property.

EXPORTING COMPARISONS

- 1 Click “↓ CSV” to download the full comparison table.
- 2 Click “↓ JSON” to export the raw record data for all queued accessions.

6 Visualiser — Charts & Plots

The Visualiser provides interactive charts for a single record or a pair of records. Select a chart type from the left sidebar, choose your record(s), and export as PNG or CSV.

BAR CHART

Shows all 20 amino acid values for one AAIndex1 record as a coloured bar chart.

- 1 Select "Bar chart" from the chart type list.
- 2 Choose a record using the filter and select list in the sidebar.
- 3 Positive values are shaded blue (darker = larger); negative values are shaded red.
- 4 Missing values render as grey bars.
- 5 Hover any bar to see the full amino acid name and exact value.
- 6 Export with "↓ CSV" (values table) or "↓ PNG" (chart image).

RADAR CHART

Displays the 20 amino acid values radially — useful for comparing the profile shape between two records.

- 1 Select "Radar chart".
- 2 Optionally tick "Overlay second record" to add a second series in green.
- 3 Values are normalised to [0, 1] per record before plotting so both series are comparable regardless of scale.
- 4 Hover the chart to see raw values per amino acid.

MATRIX HEATMAP

Renders an AAIndex2 or AAIndex3 substitution / contact potential matrix as a colour grid.

- 1 Select "Matrix heatmap".
- 2 Choose AAIndex2 (mutation matrices) or AAIndex3 (contact potentials) from the database buttons.
- 3 Select a matrix record. The 20×20 grid renders with positive scores in blue and negative in red.
- 4 Hover any cell to see the row/column amino acids and the score.
- 5 Export the full matrix as CSV.

PROPERTY SCATTER

Plots all 20 amino acids as labelled points in a 2D space defined by two AAIndex1 properties. Dashed crosshairs mark the mean of each axis.

- 1 Select "Property scatter".
- 2 Choose the X axis record from the top selector and the Y axis record from the second selector.
- 3 Each point is labelled with the single-letter amino acid code.

- 4 Hover a point for the full name and both property values.
- 5 Amino acids in the same quadrant share similar values on both properties — useful for understanding co-variation.

Tip: Try pairing hydrophobicity (KYTJ820101) on one axis with molecular weight (FASG760101) or charge (ZIMJ680104) on the other to reveal physicochemical clusters.

7 Favourites

Favourites are accessible across the entire application and persist between sessions.

- **Explorer:** Star any record card. Use “Show favourites only” to filter to starred records.
- **Record Detail:** Star or unstar the current record from its detail page.
- **Encode page:** The Favourites panel in the left sidebar expands to show your starred records. Click any to instantly select it as the active encoding property.

8 Exporting Data

LOCATION	FORMAT	CONTENTS
Explorer / Record Detail	CSV	All 20 amino acid values for one record
Explorer / Record Detail	JSON	Full record metadata + values
Explorer / Record Detail	AAIndex	Canonical AAIndex flat-file format (whole DB)
Sequence Analysis — Encoder	CSV	Per-position: position, amino acid, property value
Sequence Analysis — Window	CSV	Per-position smoothed values
Encode	CSV (long)	All sequences × all positions with property value
Encode	CSV (summary)	Per-sequence: length, valid count, mean, min, max
Comparator	CSV	All amino acids × all queued records
Comparator	JSON	Raw record data for all queued accessions
Visualiser (bar / radar / scatter)	CSV	Property values for selected record(s)
Visualiser (matrix heatmap)	CSV	Full substitution matrix
Any chart	PNG	Chart snapshot at screen resolution

9 Citation

If you use AAIndex Explorer or data derived from the AAIndex database in your research, please cite the original AAIndex publication:

Kawashima, S. and Kanehisa, M.

"AAindex: amino acid index database."

Nucleic Acids Research, 28(1): 374, 2000.

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The AAIndex database is maintained by the Kanehisa Laboratories and is freely available at <https://www.genome.jp/aaindex/>.